JAN–M–219

30 MAY 1945

JOINT ARMY-NAVY SPECIFICATION

MERCURY FULMINATE

Army Number
50-13-6C

Navy Number
51M8

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army	Navy Department
50–13–6B 17 April 1942	----------

A. APPLICABLE SPECIFICATIONS

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–A–183—Acid, Nitric (For Ordnance Use).
JAN–M–191—Mercury.

U. S. ARMY SPECIFICATIONS

4–1018—Ethyl Alcohol.
50–0–1—General Specification for Ammunition except Small Arms Ammunition.[1]
50–13–5—Trinitrotoluene (TNT).
100–2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing.

INTERSTATE COMMERCE COMMISSION SPECIFICATIONS

5—Steel Barrels or Drums.
5B—Steel Barrels or Drums.
10B—Wooden Barrels and Kegs.
17E—Steel Drums.

B. GRADE

B–1. This specification covers one grade of mercury fulminate as hereinafter specified.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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C. MATERIAL AND WORKMANSHIP

C–1. When materials used in the manufacture of mercury fulminate are procured by the Government they shall conform to the requirements of the applicable specifications listed in section A.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Appearance.*—Crystals of sparkling appearance which are white, gray, or light gray with a yellowish tint.

E–2. *Granulation.*—

	Percent
Retained on 149-micron (No. 100) U. S. standard sieve (maximum)	15
Through 74-micron (No. 200) U. S. standard sieve (maximum)	75

E–3. *Sand test.*[1]—Weight of sand crushed, minimum, 44 gm

E–4. *Mercury fulminate.*—Minimum, 98.0 percent.

E–5. *Acidity (as determined in par. F–4d).*—None.

E–6. *Insoluble matter.*—Maximum, 2.0 percent.

E–7. *Free mercury.*—Maximum, 1.0 percent.

E–8. *Chlorine.*—Maximum, 0.05 percent.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lots.*—Maximum, 5,000 pounds.

F–2. *Sampling.*—A minimum of 10 percent but not more than three of the barrels or drums in the lot shall be selected in such a manner as to be representative of the lot. By means of a wooden scoop sufficient material shall be taken from each of the bags of each selected container to yield a total sample of approximately 225 gm. Each sample shall be blended on a smooth surface by mixing with a horn or wooden spatula. Water shall be added in case the sample appears to be dry. The sample shall be spread out and divided into squares of approximately ½ inch in size. Small portions shall be taken from each square to make a primary sample of approximately 50 gm. wet weight. Each primary sample shall be placed in a bottle and labeled to show the source of the sample. The remainder of each sample should weigh approximately 175 gm. These shall be mixed thoroughly, spread out, marked into squares and a small portion removed from each until approximately 50 gm. aggregate has been removed. This composite sample shall be placed in a smooth-necked bottle with a tight-fitting stopper and labeled to show the name of the material, manufacturer, plant, contractor, order number, number of pounds in the lot, and lot number. Make all acceptance tests on the composite sample. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements of this specification.

F–3. *Inspection.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—The following tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Preparation of sample.*—Enough mercury fulminate (about 30 gm.) to allow material for making all of the following tests shall be placed on a filter paper folded in the usual manner in a 2½-inch glass

[1] Not required for Navy purchases except when specified.

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funnel and washed with 95-percent ethyl alcohol in order to remove water. The filter paper holding the mercury fulminate is placed in a small evaporating dish and dried in a sealed unit electric oven or a steam oven at 40°-50° C. to constant weight.

F-4b. *Granulation.*—

F-4b (1). *Wet method.*—Transfer a dried and weighed portion of approximately 10 gm. of the sample to a 149-micron (No. 100) U. S. standard sieve which has been placed in a porcelain evaporating dish containing sufficient water to cover the wire portion of the sieve. Shake the sieve for 10 minutes in such a manner that the mercury fulminate is below the surface of the water and any material passing through the sieve is caught on the evaporating dish. Transfer the portion of the fulminate retained on the sieve to a tared filter paper and wash twice with alcohol and once with ether. Dry the filter paper and fulminate at 50° C. for 1 hour, cool in a desiccator, and weigh. Calculate the increase in weight as percentage retained on the No. 100 sieve. Transfer the portion of the sample caught in the evaporating dish to a 74-micron (No. 200) U. S. standard sieve and sift as described above. Transfer the material caught in the evaporating dish to a tared filter paper, wash, dry and weigh as before. Calculate the increase in weight of the tared filter paper as percentage passing through the No. 200 sieve.

F-4b (2). *Alternate method for granulation (dry method).*—When mercury fulminate is sifted dry, the operation shall be performed in a location safely removed and/or shielded from other explosives, personnel or material which would be harmed by explosions. The motor of the shaking machine shall be started and stopped by a remote control switch. Assemble a No. 100 sieve on a No. 200 sieve and fit them on a pan. The sieves must be fitted with rubber gaskets. Place a 5 gm. sample of dry fulminate on the upper sieve and shake the sieve for 3 minutes on a Tyler Ro-Tap (or equal) machine. Carefully brush the material retained on the upper sieve into a weighing pan and weigh. Calculate the weight as percentage retained on the No. 100 sieve. Brush the material passing through the No. 200 sieve from the pan, weigh and calculate as percentage passing a No. 200 sieve.

F-4c. *Sand test.*—Transfer to each of five empty No. 6 blasting cap shells 0.400 gm. of trinitrotoluene (grade I, U. S. Army Specification 50-13-5) of such fineness that all of the material passes through a 177-micron (No. 80) U. S. standard sieve. Each blasting cap shall be composed of copper, gilding metal, or aluminum and shall be approximately 1.46 inches long by 0.217-inch inside diameter. With the cap held in a loading block, insert a plunger 0.20 inch in diameter in the shell and subject the trinitrotoluene to a pressure of 3,000 p. s. i. (100 pounds actual load) for 3 minutes. Remove the plunger and transfer to the shell 0.270 gm. of mercury fulminate. Insert a reinforcement cap (approximately 0.217 inch outside diameter, open at one end and a hole 0.11 ± 0.03 inch diameter at the other end) with the open end down. Insert the plunger in the shell and apply the pressure of 3,000 p. s. i. for 3 minutes. With a pin, prick the powder train in one end of a piece of miner's fuse 8 or 9 inches long. Crimp to the pricked end one of the caps loaded as directed above taking care that the end of the fuse is held firmly against the charge in the cap. Crimp near the mouth of the cap to avoid squeezing the

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charge. Pour 80.0±0.1 gm. of standard Ottawa sand which passes through a 840-micron (No. 20) U. S. standard sieve and is retained on a 590-micron (No. 30) U. S. standard sieve, into the cavity of the sand test bomb (see Fig. 1). Level the sand by striking the bomb vigorously with a hammer at least five times while rotating the bomb on its axis. Lower the fuse and cap into the bomb cavity so that the cap is centered at the axis of the bomb and just touches the sand. Hold the cap and fuse in this position and carefully pour 120.0±0.1 gm. more sand into the bomb, being careful not to disturb the position of the cap. To avoid possible error caused by the explosion blowing crushed sand through the hole in the cover, slip a piece of rubber tubing, approximately ⅛ inch long and of such inner diameter that it fits the fuse snugly over the fuse and adjust at a point on the fuse so that the rubber will be against the inner side of the bomb cover with the loaded cap in position as described above. Carefully insert the fuse through the hole in the bomb cover, lower the cover into position, and fasten securely. Strike the bomb vigorously with a hammer at least 5 times while rotating the bomb on its axis. Light the fuse, and after the explosion takes place, transfer the sand to a sheet of smooth glazed paper, taking care to remove any sand which may adhere to the sides of the bomb, or to pieces of the detonator shell, or burnt fuse. Empty all the sand on a 590-micron (No. 30) U. S. standard sieve fitted with a pan. Weigh the sand which passes through the sieve after shaking 3 minutes, using a mechanical shaker.

F–4d. *Mercury fulminate.*—Transfer a weighed portion of 0.3 gm. of the dried sample to a 250 ml. beaker.[1] Add quickly 30 ml. of a freshly prepared solution of sodium thiosulfate (20 gm. per 100 ml.). Shake the mixture for exactly 1 minute. Titrate immediately with approximately 0.10 *N* HCl using three drops of methyl red indicator. It is important that titration begin exactly 1 minute after the addition of the sodium thiosulfate. Add most of the HCl required immediately and complete the titration in not more than 1 minute. Make a blank determination with the same quantity of sodium thiosulfate solution. Calculate the percentage of mercury fulminate as follows:

$$\text{Percent mercury fulminate} = \frac{7.116(V-v)N}{W}$$

where:

V = ml. HCl required for titration
v = ml. HCl required for blank
N = normality of HCl used
W = weight of sample

F–4e. *Acidity.*—Transfer a weighed portion of 10.0 ±0.1 gm. of the dried sample to a filtering crucible provided with a filter paper disc. Wash the sample with two 25-ml. portions of boiled distilled water which has been cooled to room temperature. Add to the filtrate three drops of methyl orange or bromthymol blue indicator. Note if acid color appears.

F–4f. *Insoluble matter.*—Transfer to a 2-liter beaker a weighed portion of approximately 1 gm. of the sample. Add 1 liter of distilled

[1] In tests of fulminate for the Navy, 5 ml. of distilled water and 10 ml. of potassium iodide solution containing 3 gm. of potassium iodide may be added before the thiosulfate solution is introduced.

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water at 90° C. and stir while maintaining at a temperature of 90° C., until the material is dissolved. Filter the solution and transfer any insoluble material to a tared glass filter or crucible. Complete the solution and filtration within 1 hour. Prepare the filter or crucible by packing fine asbestos to a thickness of approximately 0.2 inch on the sintered glass plate of medium porosity, washing with hot water, alcohol and ether while applying suction, and then weighing. Wash the insoluble material on the filter with hot water, alcohol and ether, and weigh. Calculate the increase in weight as percent insoluble matter.

F–4g. *Free mercury.*—To the insoluble matter, obtained as described in paragraph F–4f, add 2 ml. of hot concentrated HCl, allowing this to remain for 1 minute prior to removal by suction. Subject the insoluble matter to this treatment three times, wash with hot water, alcohol and ether, and weigh. Repeat the procedure until the loss in weight is less than 0.0003 gm. To the residue add three successive 2-ml. portions of hot dilute nitric acid (1:1) in the same manner as with the HCl. Wash the residue with hot water, alcohol, and ether, and weigh. Calculate the loss in weight as percentage of metallic mercury.

F–4h. *Chlorine.*—Transfer a weighed portion of approximately 5 gm. of the dried sample to a filtering crucible equipped with a filter paper disc. Wash with two 25-ml. portions of distilled water at 90°–100° C. catching the filtrate in a test tube. Add to the filtrate three drops of nitric acid (specific gravity 1.4) and 10 drops of a 10-percent solution of silver nitrate. Compare the turbidity with that produced when a solution of 0.0042 gm. of pure NaCl in 50 ml. of distilled water is treated in the same manner. A greater turbidity indicates the presence of more than 0.05 percent chlorine in the sample.

F–5. *Retests.*—If the composite sample representative of the lot fails to pass the inspection tests the manufacturer shall have the option of having analysis of each primary sample made at his own expense. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements of this specification and resubmit the lot for acceptance, provided that the markings on the container are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—

G–1a. Unless otherwise specified, mercury fulminate in bulk shall contain, when packed, not less than 25-percent water, and in this wet condition shall be placed in bags of 4-ounce or heavier duck, each bag containing approximately 25 pounds dry weight of mercury fulminate. There shall be placed in each bag, and over the fulminate, a cap of the same cloth and diameter of the bag. The bag shall be tied securely. Five of these bags shall be placed in a larger bag of 4-ounce or heavier duck. This large bag shall be tied securely. The large duck bag shall contain not more than 150 pounds dry weight of mercury fulminate. This bag shall be placed in the center of a barrel or drum conforming to the requirements of Interstate Commerce Commission Specification 5 or 5B, 17E or 10B. Each container shall be lined with a heavy close fitting bag made of jute or other suitable bag material of equal strength. The large duck bag shall be completely surrounded within the jute bag by not less than 3 inches

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of well packed sawdust saturated with water. The jute bag shall be closed by secure sewing to prevent the escape of sawdust. The outer container shall be watertight.

G–1b. Unless otherwise specified, when considered necessary to prevent freezing in shipment and storage, the mercury fulminate shall be wet with not less than 25 percent (fulminate weight) of a solution of denatured ethyl alcohol and water containing not less than 50 percent by weight of ethyl alcohol. The 25-pound duck bags shall be placed in a strong waterproof bag instead of a duck bag, and a drum or barrel shall be filled with sawdust saturated with the same 50 percent by weight solution of ethyl alcohol. The waterproof bag shall withstand the action of 50 percent by weight of alcohol in water and shall have no deleterious effect on the mercury fulminate. The mercury fulminate shall not be colored or otherwise contaminated by compounds leached from the sawdust.

G–2. *Marking.*—All containers shall be marked in accordance with the requirements of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles by Freight, and unless otherwise specified in the contract or order shall be marked with the following information completed:

MATERIAL _____
LOT. NO. _____
SPECIFICATION NO. _____
MANUFACTURER _____
CONTRACTOR _____
CONTRACT NO. _____
GROSS WEIGHT _____
DATE OF MANUFACTURE _____

In addition to the foregoing, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisition, schedules, and contracts or orders should contain the title of the specification, the number, and the date.

H–2. *Use.*—Mercury fulminate covered by this specification is intended for use in detonators and primers.

H–3. Mercury fulminate is classed as a dangerous, sensitive explosive. All tests should be done in a room set aside for work on primary explosives only. Extreme precaution should be taken in all stages of the analytical procedure, and especial care must be taken to avoid accumulation of static charge. In general, no quantity of mercury fulminate should be taken larger than needed at any one time.

H–4. Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–5. Copies of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles by Freight, of which Interstate Commerce Commission Specifications 5, 5B, 10B, and 17E form a part, may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 40 cents.

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifica-

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tions may be obtained, as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications, through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-7. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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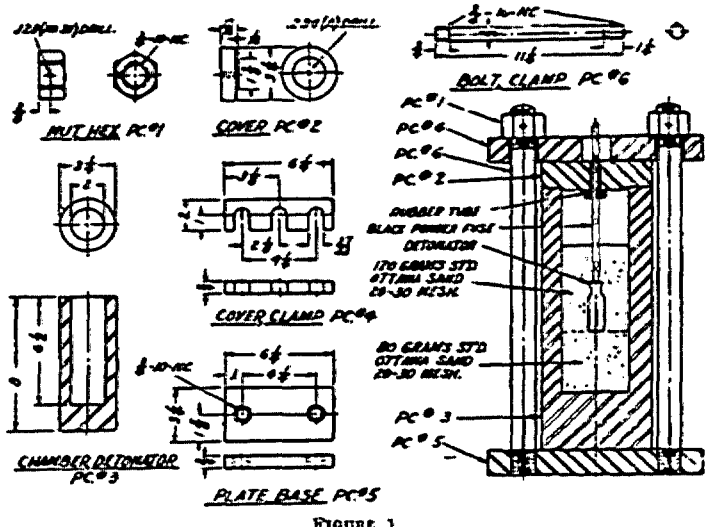


FIGURE 1.

[JAN-M-219] (no. 64)

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